

**Time:** 5 minutes**Outcome:** Calculate Cross-Price Elasticity of Demand.**Difficulty:**

Intermediate

Cross-Price Elasticity of Demand



THE CORE IDEA

Cross-price elasticity measures how demand for one good responds to a price change in another good. A large positive cross-price coefficient signals a strong substitution relationship. Complementary goods connect through opposite movements between one price and the other good's demand. Demand for one good moving in the same direction as the other good's price implies positive cross-price elasticity. Little demand response to the other good's price implies little economic relationship between the goods.



HOW TO RECOGNIZE IT

- A positive cross-price elasticity indicates substitutes; a negative value indicates complements.
- If Y's price falls 8%, a rise in demand for X indicates complements; a fall indicates substitutes.
- Use the price of the other good, not the good's own price.



WATCH OUT

Do not confuse cross price with income or own price. A negative cross-price elasticity indicates complements, so higher printer prices reduce demand for ink.



WORKED EXAMPLE: IDENTIFYING COMPLEMENTS

The price of printers rises by 10%, and demand for ink cartridges falls by 7%. Cross-price elasticity is $-7\% / 10\% = -0.7$. The negative sign indicates that printers and ink cartridges are complements because a higher price for printers reduces demand for ink.



CHECK YOURSELF

Cross-price elasticity of ink demand with respect to printer price is -0.7. If printer price falls 5%, calculate the predicted percentage change in ink demand and state its direction.

**READY?**

Return to the game and master this concept.